INTERNATIONAL MONETARY FUND

Corporate Sector Stress Testing with Endogenous Balance Sheet Dynamics

Marco Gross and Rui Xu

WP/26/209

IMF Working Papers **describe research in progress by the author(s) and are published to elicit comments and to encourage debate.**
The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

2026
OCT

WORKING PAPER

© 2026 International Monetary Fund WP/26/209

IMF Working Paper
Western Hemisphere Department

Corporate Sector Stress Testing with Endogenous Balance Sheet Dynamics

Prepared by Marco Gross and Rui Xu

Authorized for distribution by Daniel Leigh
October 2026

IMF Working Papers **describe research in progress by the author(s) and are published to elicit comments and to encourage debate.** The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

ABSTRACT: This paper presents a stress test framework for nonfinancial corporate (NFC) balance sheets. The model simulates firm income statements and balance sheets conditional on macro-financial scenarios, to obtain conventional risk metrics such as probabilities of default (PDs) and loss given default (LGD), alongside other structural metrics such as interest coverage ratios, leverage, and others. The model entails explicit stock–flow accounting and a link of borrowing costs to firm-specific credit risk metrics, i.e., an endogenous cost of funding feedback. An illustration is provided with an application to Brazil, involving 8,000 private firms and three hundred listed nonfinancial firms. The results point to heightened vulnerabilities under stagflation shock scenarios, which jointly weaken earnings and raise funding costs, increasing default risk and debt at risk.

RECOMMENDED CITATION: Gross, M., and Xu, R. 2026. “Corporate Sector Stress Testing with Endogenous Balance Sheet Dynamics.” IMF Working Paper 26/209, International Monetary Fund, Washington DC.

JEL Classification Numbers:	G3, G21, G17, M41, E43.
Keywords:	Corporate sector credit risk, stress tests, dynamic balance sheets, probability of default, loss given default.
Author’s E-Mail Address:	mgross@imf.org, rxu@imf.org.

WORKING PAPERS

Corporate Sector Stress Testing with Endogenous Balance Sheet Dynamics

Prepared by Marco Gross and Rui Xu[1]

[1] We are grateful to Central Bank of Brazil and IMF seminar participants for their helpful comments and suggestions. All errors are solely the authors' responsibility.

Contents

I. INTRODUCTION 5

II. MODEL FRAMEWORK 6

III. APPLICATION TO BRAZIL 11

IV. CONCLUSIONS 14

REFERENCES 15

ANNEX

I. DATA COLLECTION AND PROCESSING 25

FIGURES

1. Model Logic—Schematic Overview 6
2. Frye-Jacobs Model—Available Data and Imputations 8
3. Macrofinancial Context 17
4. Nonfinancial Corporates Financing 18
5. Capital IQ Sample 19
6. PD and LGD Models 20
7. Macro-Financial Scenarios 21
8. NFC Stress Testing Results 22
9. NFC Stress Testing Results 23
10. The Role of Risk Premia 24

I. Introduction

Nonfinancial corporations (NFCs) are central to macro-financial stability, given their large share in aggregate credit exposures and their interconnectedness with banks, nonbank financial institutions, and capital markets. Adverse macro-financial shocks can weaken firms' profitability, liquidity, and debt servicing capacity, transmitting stress to the broader financial system through rising credit losses, tighter financial conditions, and reduced investment activity. For this reason, corporate sector stress testing has been a critical component of systemic risk surveillance by central banks and international financial institutions, including in the IMF's Financial Sector Assessment Programs (FSAPs). Recent FSAP applications—including Italy, the United Kingdom, Japan, and Brazil—have shown the need for methodologies capable of linking macro-financial scenarios to firm-level balance sheet dynamics and credit risk outcomes.

Existing approaches to corporate stress testing nevertheless are subject to three important limitations. First, many studies rely on static balance sheets, isolated shocks, or reduced-form regressions, without dynamically simulating firm income statements and balance sheets under coherent macro-financial scenarios (e.g., Chow, 2015; Chan-Lau et al., 2017; IMF, 2020; IMF, 2022; Central Bank of Brazil, 2025). Second, even approaches that model balance sheet dynamics typically do not incorporate endogenous risk premia into firms' borrowing costs, despite the fact that credit spreads can rise sharply during stress episodes (models without such feedback include, for example, Tressel and Ding, 2021 and Caceres et al., 2020). Third, existing NFC stress testing frameworks usually do not project firm-level loss-given-default (LGD), and—in that context—their outputs are often not integrated into broader systemic risk exercises such as bank solvency stress testing, due to missing probability of default (PD) /LGD outputs. In practice, LGD modeling is frequently conducted only at the aggregate bank portfolio level rather than at the firm level.

Against this background, this paper develops a structural framework for corporate sector stress testing with endogenous balance sheet dynamics. The framework combines firm-level stock-flow accounting with macro-financial scenarios to simulate the evolution of income statements, balance sheets, and credit risk metrics over time, for a potentially large sample of individual nonfinancial firms. Borrowing costs are modeled endogenously with firm-specific risk premium linked to projected PDs and LGDs, i.e., two-way feedback is implemented and shown to matter empirically. PDs are estimated using a firm-level panel model rooted in Merton-model rationale, while LGDs are projected using the structural framework of Frye and Jacobs (2012). The framework therefore jointly captures the interaction between macro-financial conditions, firm cash flows, leverage, debt servicing capacity, and credit risk.

The framework is designed not only as a corporate vulnerability analysis tool, but also as an operational input into broader systemic risk assessments. By producing firm-level and segment-level PD and LGD projections, the outputs of the model can serve as inputs to bank solvency stress testing exercises and climate risk analysis frameworks. This has been demonstrated in applications such as the ENV-FIBA model (Gross et al., 2025), where the micro simulation layer for NFCs is similar to the one we present in this paper. The structural micro data-based model approach is particularly useful in settings where historical default data are sparse, where observed PD variation is limited, or where structural interpretation of stress transmission channels is important for policy analysis. The methodology was implemented in the Japan FSAP and further refined for the Brazil FSAP, illustrating its applicability across jurisdictions and macro-financial environments.

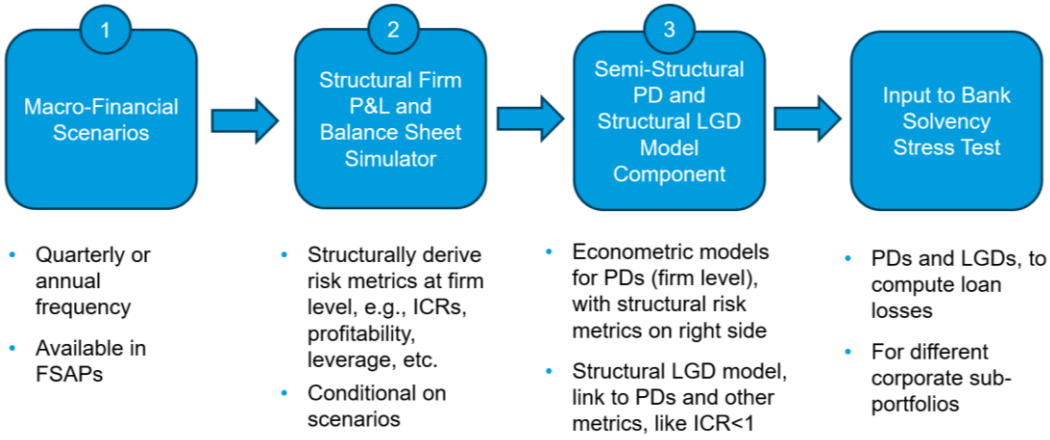
The remainder of the paper is structured as follows. Section II presents the model framework, including all its components. Section III applies the methodology to Brazil, using firm-level data from Capital IQ and CreditEdge, conditional on FSAP stress scenarios. Section IV concludes.

II. Model Framework

Overview

The schematic in Figure 1 summarizes the logic of the model. Macro-financial scenarios form the starting point, which are constructed at quarterly or annual frequency, and usually involve a baseline and one or more adverse scenarios with a three-five-year horizon. The structural firm balance sheet simulator—the second component in Figure 1—simulates the individual firms' profits and losses (P&L) and balance sheets. Structural risk metrics are the outcome of this module, including metrics such as interest coverage ratios, profitability metrics, liquidity risk metrics, and others, over the scenario horizon. In Module 3—the semi-structural PD and structural LGD module—these firm-level risk metrics are translated to PDs and LGDs. Finally, an "off-model component" at the end entails an optional link to bank solvency stress testing, where the PDs and LGD paths can be used to compute loan losses from the corporate portfolios for banks. In the following sub-sections, the presentation of the core of the model will be shown in reverse order to see what inputs are required for the desired outputs.

Figure 1. Model Logic—Schematic Overview



Note: The schematic summarizes the model components and model flow. Modules 2 and 3 represent the core of the model. They are described in detail in Section II.

PD Model Component

An econometric PD model component is estimated using a sample of firms for which financial statements and historical default rates are both available. The structure of the firm-panel regression in eq. (1) is informed by a stock-and-flow-based Merton model rationale. That is, it relates default rates (historical PDs) to drivers such as leverage (a stock-stock ratio), interest coverage ratio (a flow-flow ratio, the ratio of net income before tax over interest expenses), and various other metrics on solvency, liquidity, and debt service ability. The dependent

variable is a logit transformation of PD, converting probability into log-odds so that linear panel regression can be applied. This transformation guarantees that the projected PD trajectory remains within the 0–1 interval.

Macro-financial variables may be included in the equation, in addition to the firm level metrics. However, in principle, the macro-financial variables feed structurally through the firm level metrics, and their separate inclusion may not be needed. For example, interest rate paths will be linked specifically to the denominator of the ICR. Notwithstanding this structural argument, from an empirical perspective, the direct inclusion of additional macro-financial controls might be warranted, however, to improve the predictive performance of the model. Other adjustments can be made to the PD model to improve predictive performance. The equation can be structured as follows:

$$logit(PD_{ft}) = \beta_0 + \beta_1 LEV_{ft} + \beta_2 ICR_{ft} + \beta_3 \mathbb{I}(ICR < 1)_{ft} + \beta_4 CDR_{ft} + \beta_5 ROA_{ft} + \beta_6 \log(Asset_f) + \beta_7 RGDP\mathrm{g_t} + \beta_8 PR_t + \beta_9 I_{ind} + \varepsilon_{ft} \quad (1)$$

where LEV_{ft} is the leverage of firm f at year t measured by total debt over total assets; ICR_{ft} is the interest coverage ratio defined as earnings before interest and taxes (EBIT) over net interest expenses; $\mathbb{I}(ICR < 1)_{ft}$ is an indicator that takes value 1 if ICR is smaller than 1 and 0 otherwise; CDR_{ft} is the cash to debt ratio; ROA_{ft} is the return on asset; and $\log(Asset_f)$ is the logarithm of average assets of firm f in the last three years to measure its size. Macro variables include real GDP growth $RGDP\mathrm{g_t}$ and monetary policy rate PR_t. Industry fixed effects are also included.

The estimated model will be used for two purposes. First, to impute PDs for firms not covered by a commercial database that holds the historical PDs for listed firms, i.e., to derive the historical PDs for unlisted firms. And second, to forecast firm-level PD paths under different scenarios for the whole sample, including listed and unlisted firms.

LGD Model Component

Loss given default (LGD) is a second principal component of credit risk analysis, as it determines the share of an exposure that is ultimately not recovered after a borrower defaults. Together with PDs and exposures at default, LGDs shape expected credit losses, bank provisioning needs, and stress test outcomes.
Yet modeling LGDs is inherently difficult. Historical LGD data are typically scarce, fragmented, and often unavailable at the firm level, particularly outside supervisory datasets. Moreover, a precise assessment of LGD in principle requires accounting for a wide range of factors, including collateral composition and valuation, legal claim structures, seniority rankings, recovery procedures, and the liquidation value of assets under stressed market conditions. Recovery rates tend to be state-dependent, deteriorating during systemic downturns when asset prices fall and insolvency processes become congested. These complexities make structural and parsimonious approaches to LGD modeling particularly relevant in stress testing applications.

The literature has documented that PDs and LGDs tend to co-move systematically over the credit cycle (see Frye, 2000; Altman and Karlin, 2010). Periods of elevated default risk are typically associated with weaker recovery values, reflecting falling collateral prices, distressed asset sales, and congested insolvency procedures during downturns. This structural relationship makes PDs a natural anchor for modeling LGDs in stress testing frameworks.

Against this background, we adopt a parsimonious yet robust methodology proposed by Frye and Jacobs (2012) in our model. Their framework assumes that default rates and loss rates are jointly driven by common systematic risk factors in an asymptotic portfolio setting. Under a Vašíček distribution for credit losses, point-in-time LGDs can be derived as a function of projected PDs and a calibrated LGD risk index, providing a tractable and economically intuitive way to generate scenario-consistent LGD paths at the firm level.[2]

$$\mathrm{LGD_t} = \frac{\Phi(\Phi^{-1}(\mathrm{PD_t}) - \mathrm{k})}{\mathrm{PD_t}} \tag{2}$$

$$\text{where } \mathrm{k} = \frac{\Phi^{-1}(\mathrm{PD}) - \Phi^{-1}(\mathrm{PD} \times \mathrm{LGD})}{\sqrt{1-\rho}} \tag{3}$$

$\mathrm{LGD_t}$ is a point-in-time (PiT) LGD at time t; $\mathrm{PD_t}$ is the PiT PD at time t; k is referred to as the LGD risk index, which is a function of Through-The-Cycle (TTC) PD and TTC LGD, with lower k indicating higher risk, and ρ is an asset correlation parameter which can be estimated from micro data or set using judgement.[3]

While PiT LGD can be obtained from CreditEdge and central banks, TTC LGD is hard to obtain in the data. Instead, the firm-specific risk index k can be calibrated to match observed PiT PD and PiT LGD.

The Frye-Jacobs methodology can be implemented flexibly depending on which combination of PiT and TTC PD and LGD are available for the individual firms (or at sectoral level). As shown in Figure 2, if one of the three variables—PD PiT, PD TTC, or LGD TTC—is not available, the "open" model parameter can be calibrated to match the initial LGD PiT, conditional on the LGD model structure in equations. (2) and (3). When all four parameters are available, no additional calibration step would be required.

The applications to Brazil and Japan fall under Case 2. Firm-level PDs were available for listed companies and estimated for unlisted firms over the sample period, providing both PiT and TTC PD. Point-in-Time LGD was provided at the industry level, and each firm's LGD risk index k is calibrated using equations (2) and (3).

Figure 2. Frye-Jacobs Model—Available Data and Imputations

	Risk Parameters Available in T0				Comments
	PD TTC	LGD TTC	PD PiT	LGD PiT	
Case 1		x	x	x	Shift the respectively unavailable T0 parameter such that the model matches the T0 LGD PiT.
Case 2	x		x	x	
Case 3	x	x		x	
Case 4	x	x	x		No T0 calibration step required.

Note: The table summarizes the different ways the Frye-Jacobs methodology can be used.

[2] The Basel Committee on Banking Supervision (BCBS) chose the closed-form single risk factor Vasicek model for regulatory capital calculations. As shown in Milonas and Vuuren (2024), the Vasicek distribution is a good approximation for expected credit losses in large, well-diversified portfolios, but it may fail to accurately capture extreme unexpected losses in the tail of the distribution.

[3] Through-the-cycle (TTC) PD estimates long-term default risk averaged across economic cycles, while Point-in-time (PiT) PD measures default risk based on current economic conditions. TTC PD estimates can be obtained by averaging the PiT PD over time.

Firm-Level Simulation

Corporate balance sheets and income statements can be obtained from commercial data vendors or administrative data sources from country authorities. Commercial data sources, such as Moody's Orbis and S&P Capital IQ, have been widely used. In some cases, such as the FSAPs for Denmark and Colombia, confidential data from country authorities have been used (IMF, 2020; IMF, 2022). Databases with broader coverage and timely updates are preferred for a more representative NFC sample and less need for historical data imputation. The raw data should be cleaned before the analysis (see Annex I for the data cleaning procedures conducted in the Brazil application).

Historical PD estimates for publicly listed firms are obtained from CreditEdge (Moody's Analytics), which provides one-year-ahead Expected Default Frequency (EDF) measures at daily frequency. These industry-standard indicators are calibrated and validated against observed default events and credit spread dynamics. EDF estimates are derived from a structural credit risk framework that incorporates three key inputs: the market value of firm assets, asset volatility, and the firm's capital structure. To align the PD data with the frequency of the Capital IQ balance sheet information, we use annual averages of the daily EDF series.
CreditEdge also provides LGD estimates for publicly listed firms. Unlike EDF measures, which are firm-specific, the LGD estimates are provided at a more aggregated level, differentiated across thirteen economic sectors, three geographic regions, and two levels of debt seniority. The estimates are derived from market spread information and recovery assumptions embedded in credit pricing.

Central banks may also estimate PDs and LGD parameters using supervisory micro data. When available, such information can complement the CreditEdge dataset, whose coverage is largely limited to publicly listed firms and larger corporates. Ideally, PD estimates would be available at the individual firm level. In practice, however, such granular supervisory estimates are often not accessible for stress testing purposes due to confidentiality constraints and data-sharing limitations, or simply unavailable. We therefore rely on the econometric PD model framework to impute PDs for firms not covered by CreditEdge, using firm-level financial characteristics and macro-financial variables.[4]

We project firm financial income dynamics using accounting identities. Firms' net income, in flow terms, evolves as follows:

$$\text{Net Income (NI)} = \underbrace{\text{Revenues - Cost of Goods Sold- Other Operating Expenses}}_{\text{depends on real GDP growth and inflation}} - \underbrace{\text{Net Interest Expenses}}_{\text{depends on policy rate, debt, and lagged PD and LGD}} - \underbrace{\text{Taxes}}_{\sim EBT} \quad (4)$$

Firm-level sales revenues are linked to real GDP growth and consumer price inflation through empirically estimated panel regressions with firm fixed effects. Cost of goods sold (CoGS) are assumed to evolve proportionally with revenues, while the sensitivity of other operating expenses to macroeconomic conditions is likewise estimated econometrically. Taxes are modeled as a constant share of earnings before taxes whenever profits are positive, and set to zero when earnings turn negative. The evolution of net interest expenses, which constitutes the key endogenous financial feedback channel in the framework, is discussed in detail below. The

[4] Imputing the PD of unlisted firms using PD model estimated from listed firms could introduce extrapolation bias. Such bias can be mitigated with better PD estimates for unlisted firms from administrative data sources.

links represent the structural connection of the macro-financial scenario inputs (Module 1 in Figure 1) to the structural firm simulator (Module 2 in Figure 1).

Net income dynamics would inform stock variables in the balance sheets, including total assets, total debt, total equity, and cash (and equivalents) stock. The change depends on net income and whether new debt is issued.

- If net income is positive, we assume that firms do not distribute dividends and instead retain all earnings on the balance sheet.[5] Of retained profits, 10 percent is allocated to the accumulation of cash buffers, while the remaining 90 percent is reinvested into the firm, consistent with the assumption that revenues expand alongside economic growth. Total debt is assumed to remain constant over time, with maturing liabilities rolled over through new borrowing.
- If net income turns negative, firms are assumed to absorb losses by drawing down existing cash buffers. Once cash reserves are depleted, firms are assumed to issue additional debt to finance the remaining funding shortfall and maintain operational continuity.[6]

The equations below show the evolution of key stock variables discussed above:

$$Total\ Assets_{ft} = Total\ Assets_{f,t-1} + NI_{ft} \underbrace{-\mathbb{I}\left(NI_{ft} + Cash_{t-1} < 0\right) \cdot (NI_{ft} + Cash_{t-1})}_{Additional\ Debt\ if\ cash\ not\ enough\ to\ cover\ negative\ NI} \tag{5}$$

$$Total\ Debt_{ft} = Total\ Debt_{f,t-1} \underbrace{-\mathbb{I}\left(NI_{ft} + Cash_{t-1} < 0\right) \cdot (NI_{ft} + Cash_{t-1})}_{Additional\ Debt\ if\ cash\ not\ enough\ to\ cover\ negative\ NI} \tag{6}$$

$$Cash_{ft} = Cash_{f,t-1} + \underbrace{\mathbb{I}\left(NI_{ft} \geq 0\right) \cdot 0.1NI_{ft}}_{when\ NI\ is\ positive} + \underbrace{\mathbb{I}\left(NI_{ft} < 0\right) \cdot \mathbb{I}\left(NI_{ft} + Cash_{f,t-1} \geq 0\right) \cdot NI_{ft}}_{when\ negative\ NI\ can\ be\ fully\ covered\ by\ cash\ stock}$$

$$\underbrace{-\mathbb{I}\left(NI_{ft} + Cash_{f,t-1} < 0\right) \cdot Cash_{f,t-1}}_{Deplete\ cash\ stock\ if\ NI\ is\ too\ negative} \tag{7}$$

Net interest income for firms has an interest income and expense component. Interest income is held constant, for simplicity and conservatism. Interest expenses, on the other hand, represent a core endogenous model component. Interest expense depends on the total debt stock and interest rates. The total debt stock dynamics were outlined above (eq. 6). The firm-specific borrowing cost i_{ft} depends on policy rates and risk premium, derived from a bank loan pricing equation:

$$\underbrace{PD_{ft} \times LGD_{ft}}_{\text{Expected Loss}} + \underbrace{\left(1 - PD_{ft}\right) i_t^{COF}}_{\text{Expected Interest Expense}} = \underbrace{(1 - PD_{ft}) i_{ft}}_{\text{Expected Interest Income}}$$

$$\leftrightarrow \quad i_{ft} = i_t^{COF} + \underbrace{\frac{PD_{ft} \times LGD_{ft}}{1 - PD_{ft}}}_{Risk\ premium} \tag{8}$$

[5] This simplifying assumption may under-estimate leverage for some firms and can be relaxed for future applications. It does not affect the PD and Debt-at-risk under stress scenarios where firms tend to have negative net income.

[6] For simplicity, the framework assumes that firms rely on cash buffers to deal with adverse shocks. This can best illustrate the effect of macro and interest rate shocks on firms' performance. In practice, firms may react by cutting investment, selling assets, renegotiating debt and other strategies. These can be easily adapted by the framework.

where i_t^{COF} is the banks' cost of funding that moves with policy rates; the firm-specific risk premium depends on both PD and LGD; i_{ft} is the interest rate for new debt or floating-rate existing debt for firms.[7]
Financial statements of firms typically do not contain detailed maturity ladder of debt, origin of the debt (domestics vs. external) or the information on whether the debt has floating or fixed interest rates. As a result, we cannot model the exact share of debt that will be subject to new interest rates in each period. To get around this issue, we empirically estimate the sensitivity $\beta^{f(i,j)}$ of average interest rate to policy rate (r_t) by sector and size using historical data.

With the sensitivity of firm borrowing rates to base rate changes, we can devise the final equation to project firm-level cost of debt. Equation (9) governs the evolution of firm level borrowing rates, anchored in a firm's observed "T0" borrowing rate, forward in time, as a function of changes in policy rates in the scenario and the endogenously evolving PDs and LGDs of each individual firm. PDs and LGDs, in turn, are a function of interest expenses, through the ICR metric in eq. (1).[8]

$$\Delta i_{f,t+1} = \beta^{f(i,j)} \Delta r_{t+1} + \Delta \left(\frac{PD_{f,t} \times LGD_{f,t}}{1 - PD_{f,t}} \right) \quad (9)$$

III. Application to Brazil

Context

Following a vigorous post-pandemic rebound, Brazil's growth is now slowing as stricter policies take effect. The country's recovery has been exceptional compared to global trends, with real GDP returning to its pre-pandemic growth trajectory. Between 2022 and 2024, real GDP grew by an average of 3.1 percent, fueled by fiscal stimulus, unprecedented agricultural production, and increased oil output. Inflation surged in 2024 due to robust consumer demand, rising food costs, and currency depreciation. In response, the central bank began tightening monetary policy in September 2024, raising the policy rate to 15 percent by June 2025 (Figure 3). This action, along with scaled-back fiscal support, curbed inflation but slowed GDP growth to 2.3 percent in 2025. Unemployment dropped to its lowest in twenty years, reaching 5.1 percent by December 2025.

NFCs in Brazil obtain credit from domestic banks and from fixed income markets in roughly equal shares (Figure 4). While external credit has remained at around 17 percent of GDP since 2017 with most of it intercompany loans, the composition of domestic credit has changed significantly in the last ten years. Earmarked credit for firms has decreased notably, dropping from 14 percent of GDP in 2015 to 8 percent in 2025, primarily as a result of the 2018 reform of the national development bank (BNDES), which brought lending rates closer to policy rates. The contraction has had a more pronounced impact on MSMEs compared to large firms, as larger firms have been able to issue bonds. The total stock of domestic debt securities increased from 6 percent to 19 percent of GDP over the past decade.

[7] The model here focuses on the price channel of financial tightening. Banks may also adjust quantity under stress scenarios through credit rationing, shorter maturities, and tighter collateral requirements. The quantity response can be added in future research.

[8] To avoid a two-way time contemporaneous relationship in the two dimensions, we lag one relation, in eq. (9), by having PDs and LGDs dated time *t*, while the dependence of PDs on interest rates (via the ICR) in eq. (1) is time contemporaneous. In this way, we circumvent the need for embedding solution algorithms, such as Gauss-Seidel.

Both demand and supply factors have contributed to the expansion of the domestic bond market (Figure 4). Investors show a preference for bonds over equities in a high-interest rate environment after 2022, while firms favor bond issuance over bank loans due to lower spreads—particularly those issuances that exempt investors from income taxes. Most debentures are issued by large companies within the services and utilities sectors; typically featuring floating interest rates. This trend has raised leverage and increased the susceptibility of these companies to rising interest rates.

Corporate profits have improved since 2018 amid strong economic growth, but elevated interest rates and higher leverage are straining corporate balance sheets. A record 2466 companies filed for judicial reorganization in Brazil at the end of 2025, doubling the number in 2022.[9] Agriculture and Services account for most companies in recovery, reflecting greater exposure to volatile cycles. Number of firms with delinquent debt has also gone up significantly in 2025 (Figure 3).

Brazilian Data

For Brazil, firm-level financial statements were obtained from Capital IQ Pro over the period 2012–2024 (Figure 5). The sample comprises approximately 8,000 private firms and three hundred publicly listed firms, collectively representing more than 70 percent of total corporate debt and around 50 percent of activity in industrialized sectors. Based on Brazil's official size classification thresholds, roughly half of the firms in the sample are medium-sized enterprises, one third are large corporates, and the remainder are small and micro enterprises (SMEs).

PD estimates are obtained from CreditEdge, which covers approximately five hundred large firms that are either publicly listed or active in bond markets. The data exhibit clear cyclical behavior, with PDs rising during periods of economic stress, including the 2009 global financial crisis and Brazil's 2015–16 domestic recession. While the pandemic episode did not trigger a comparable spike in PDs—largely reflecting extensive policy support measures—the subsequent monetary tightening cycle and higher interest rates contributed to a renewed increase in corporate default risk during 2022–24.

PiT LGD estimates were obtained from both CreditEdge and the Brazilian Central Bank (BCB), with both sources providing LGD measures at the sectoral level. For firms not covered by CreditEdge, we rely on the BCB's sector-specific LGD estimates for 2025 as calibration inputs.

PD Model Estimates for Brazil

As CreditEdge primarily covers publicly listed firms and large bond issuers, we estimate PDs for the remaining roughly 8,000 firms in the Capital IQ sample using the econometric framework introduced in Section II. Equation (1) was estimated on the subset of firms for which CreditEdge PD observations are available. The estimated coefficients are statistically significant and economically consistent with theoretical priors, as shown in Figure 6. The resulting model is then used to impute PDs for firms outside the CreditEdge universe. The median imputed PD series displays levels and cyclical dynamics broadly consistent with supervisory PD estimates produced by the BCB using loan-level data, supporting the robustness of the imputation.

[9] Judicial reorganization is a formal process in which the company presents a restructuring plan to all creditors, under the supervision of a judicial administrator and following strict legal deadlines.

Results for Brazil

We project balance sheets and cash flows of Brazilian firms in the Capital IQ sample over the period 2025–2028 under three macro-financial scenarios: a baseline, a recessionary scenario, and a sovereign debt distress scenario, as illustrated in Figure 7.[10] The sovereign debt distress scenario generates the most severe pressures on corporates because it combines recessionary dynamics with inflationary forces. A loss of fiscal confidence triggers currency depreciation and higher inflation, prompting the inflation-targeting central bank to tighten monetary policy despite weakening economic activity.

The results indicate that Brazilian corporates remain vulnerable to stagflation shocks despite entering the stress horizon with relatively strong profitability (Figure 8). Under the sovereign debt distress scenario, borrowing costs rise sharply as both policy rates (SELIC) and firm-specific risk premia increase. At the same time, leverage ratios deteriorate as firms draw down cash buffers and accumulate additional debt to absorb negative net income. Profitability metrics also weaken materially: the EBIT-to-assets ratio declines due to lower sales and weaker economic activity, while return on assets falls further as higher interest expenses compress net earnings. The share of firms with low interest coverage ratios (ICR<1)—indicating insufficient operating earnings to cover interest payments—increases significantly, although it remains below the peaks observed during the 2015–16 crisis. Nevertheless, aggregate debt at risk, defined as debt held by firms with ICR below one, is projected to exceed 2015–16 levels owing to the substantially higher leverage of large firms. PDs and LGDs likewise rise notably, approaching levels observed during the previous domestic crisis episode.

The results also reveal heterogeneities across firm size segments, with implications for the calibration of corporate credit risk in bank solvency stress tests. Supervisory data used in stress testing exercises typically distinguish between exposures to large corporates and to micro, small, and medium-sized enterprises (MSMEs). MSMEs generally face higher borrowing costs, reflecting both higher default risk and a stronger reliance on bank financing, which is typically more expensive than bond market funding (Figure 4). The projected median PDs and LGDs for large corporates and MSMEs can therefore be mapped to banks' portfolio exposures to these segments, allowing for a more granular estimation of expected credit losses in the solvency stress test framework (IMF, 2026).[11]

Corporate vulnerabilities also differ markedly across economic sectors (Figure 9). Mining and utilities benefit from the lowest borrowing costs, reflecting their access to earmarked credit, tax-advantaged bond financing, and stronger market access, although they also exhibit comparatively high leverage due to their capital-intensive business models. These sectors generally maintain stronger profitability and lower default risk under stress scenarios. Construction and real estate firms display the weakest profitability and the highest PDs under both the baseline and sovereign debt distress scenarios. Manufacturing firms remain among the most profitable but face relatively elevated borrowing costs because they rely more heavily on commercial bank financing.

[10] The *recessionary scenario* features a synchronized global economic slowdown, widening credit spreads, term premium decompression, and a confidence loss that dampens aggregate demand. The *sovereign debt distress scenario* features the same global layer as the recessionary scenario, but features a confidence crisis element, pertaining to Brazil's fiscal position, leading to a risk-off event. This results in a sharper FX depreciation and even higher risk premium. Inflation increases and policy rates hike as a result. For details of the macro-financial scenarios design, please refer to IMF, 2026; and Vitek 2018.

[11] Such mapping assumes that banks' exposure to corporates have the same PD as the overall corporate PD. This may not be the case for large corporations that have a significant portion of liabilities in debentures, external debt and other capital market instruments. Detailed data on liability structure, which could be available to supervisors, can be used to supplement the financial statement data to provide an exact mapping.

Service sector firms account for more than half of aggregate debt at risk under both scenarios, reflecting both the sector's heavy weight in the economy and its comparatively weaker debt servicing capacity.

Incorporating endogenous risk premium materially changes the assessment of firms' debt servicing capacity under adverse scenarios. Much of the existing literature projects borrowing costs primarily through policy rate pass-through, thereby neglecting the sharp increase in credit risk premium that typically accompanies periods of financial stress. Our simulations indicate that risk premia account for more than half of the increase in borrowing rates under the sovereign debt distress scenario, with particularly strong effects for MSMEs (Figure 10). Ignoring this channel would substantially understate corporate vulnerabilities: the share of firms with interest coverage ratios below one would be underestimated by around 25 percent for large corporates and 30 percent for MSMEs. These results underscore the importance of incorporating endogenous credit risk premia when evaluating corporate balance sheet resilience and debt servicing capacity under stress.

IV. Conclusions

This paper develops a corporate sector stress testing framework that combines dynamic firm-level balance sheet simulation with endogenous credit risk feedback. The methodology links macro-financial scenarios to the evolution of firms' income statements, balance sheets, borrowing costs, and credit risk metrics through explicit stock-flow accounting relationships. In contrast to standard approaches based on static balance sheets or aggregate econometric models, the framework directly projects firm-level probabilities of default (PDs) and loss-given-default (LGDs), allowing the results to be integrated into bank solvency stress tests and broader systemic risk analysis.

The application to Brazil highlights the vulnerability of corporates to stagflation stress scenarios. Although firms entered the stress horizon with relatively strong profitability, a deep recession combined with higher interest rates triggered by a fiscal confidence shock would materially weaken earnings, increase leverage, and raise default risk toward levels observed during the 2015–16 crisis. Elevated initial policy rates and higher leverage among large corporates—partly driven by increased bond issuance—amplify these vulnerabilities. While MSMEs and construction firms are particularly sensitive to tighter financial conditions, large corporates and service sector firms account for the largest share of aggregate debt at risk and therefore pose the most significant systemic exposure for the banking sector.

The framework remains data intensive and subject to several limitations that point toward avenues for future work. Commercial datasets often lack detailed information on debt composition, including maturity structures and the split between bank loans and bond financing, requiring approximations through estimated interest rate pass-through elasticities. Coverage and data quality of SMEs are also limited because commercial providers rely primarily on published financial statements. In jurisdictions where SME lending represents a large share of banks' credit exposures, supervisory or central bank data may therefore need to complement the framework through econometric models based on portfolio-level PDs, NPLs, or related credit risk indicators.

References

Altman, Edward I., and Brenda J. Kuehne. "Defaults and Returns in the High-Yield Bond and Distressed Debt Market: Review and Outlook," 2014, *Palgrave Macmillan Books*, in: Torben Juul Andersen (ed.), Contemporary Challenges in Risk Management, chapter 8, pages 203-253.

Banerjee Ryan Niladri and Boris Hofmann. “The Rise of Zombie Firms: Causes and Consequences,” *BIS Quarterly Review*, 2018.

Caceres, Carlos, Diego Cerdeiro, Dan Pan, and Suchanan Tambunlertchai. "Stress Testing U.S. Leveraged Corporates in a Covid-19 World.", *IMF Working Paper 20/238*, 2020, International Monetary Fund.

Chan-Lau, Jorge A., Cheng Hoon Lim, Jose Daniel Rodríguez-Delgado, Bennett W. Sutton, and Melesse Tashu. "Bottom-up Default Analysis of Corporate Solvency Risk: An Application to Latin America", *IMF Working Paper 17/133*, 2017, International Monetary Fund.

Central Bank of Brazil, "Financial Stability Report", 2025 April.

Chow, Julian. “Stress Testing Corporate Balance Sheets in Emerging Economies,” *IMF Working Paper 15/216*, 2015, International Monetary Fund.

Frye, Jon and Michael Jacobs Jr., "Credit Loss and Systematic Loss Given Default," *Journal of Credit Risk*, 2012, 8(1), 109-140.

Frye, Jon. "Depressing recoveries", *Risk*, 2000, 13(11), 108–111.

Frye, Jon. "Loss Given Default as a Function of the Default Rate." *Federal Reserve Bank of Chicago*, 2013, 1-15.

Gross, Marco, Dimitrios Laliotis, Mindaugas Leika, and Pavel Lukyantsau. “Expected Credit Loss Modeling from a Top-down Stress Testing Perspective,” *IMF Working Paper 20/111*, 2020, International Monetary Fund.

Gross, Marco, Jinhyuk Yoo, Hugo Rojas-Romagosa, Zulma Barrail, Salim Dehmej, and Hannah Sheldon. "The ENV-FIBA Model for Climate Risk Analysis", *IMF Working Paper 25/230*, 2025, International Monetary Fund.

Gross, Marco and Javier Población. "Implications of Model Uncertainty for Bank Stress Testing," *Journal of Financial Services Research*, 2019, 55(1), 31-58.

International Monetary Fund, "Technical Note on Risk Analysis", *Colombia: Financial Sector Assessment Program*, 2022.

International Monetary Fund, "Technical Note on Systemic Risk Analysis and Stress Testing", *Japan: Financial Sector Assessment Program*, 2024

International Monetary Fund, "Technical Note: Financial Stability and Stress Testing of the Banking, Insurance, and Non-financial Corporate Sectors", *Denmark: Financial Sector Assessment Program*, 2020.

International Monetary Fund, "Technical Note: Systemic Risk Analysis and Stress Testing of the Banking and Corporate Sectors", *Italy: Financial Sector Assessment Program*, 2020.

International Monetary Fund, "Technical Note: Systemic Risk Analysis and Stress Testing", *Brazil: Financial Sector Assessment Program*, 2026

International Monetary Fund, "Technical Note: Systemic Risk Analysis and Stress Testing", *Switzerland: Financial Sector Assessment Program*, 2025.

International Monetary Fund, "Technical Note: Systemic Stress, and Climate-related Financial Risks: Implications for Balance Sheet Resilience", *United Kingdom: Financial Sector Assessment Program*, 2022.

Merton, Robert C. "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates" *Journal of Finance*, 1974, 29 (2): 449–470.

Milonas, Natasa, and Gary van Vuuren, “Simulating Credit Loss Distributions: Empirical Versus the Vasicek Model.” *International Journal of Economics and Financial Issues*, 2024, *14*(2), 77–88.

Moody's Analytics. "CreditEdge from Moody's KMV".

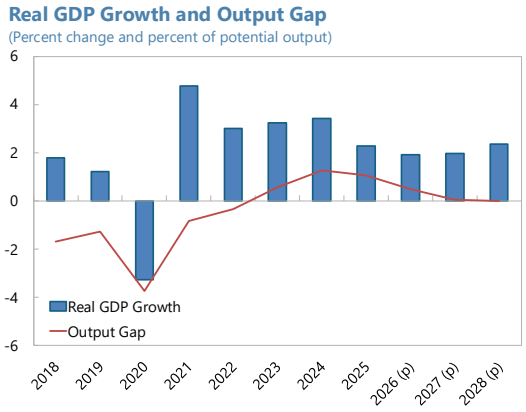
Roulet, Caroline. "Corporate Debt Stress Testing: A Global Analysis of Non-financial Corporations." *OECD Working Papers on Finance*, Insurance and Private Pensions, 2020.

Tressel, Thierry, and Xiaodan Ding. "Global Corporate Stress Tests—Impact of the COVID-19 Pandemic and Policy Responses". *IMF Working Paper 21/212*, 2021, International Monetary Fund.

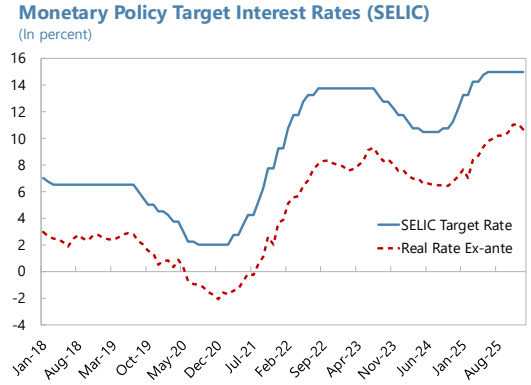
Vitek, Francis. "The Global Macrofinancial Model," *IMF Working Papers 18/081*, 2018, International Monetary Fund.

Figure 3. Brazil: Macro-financial Context

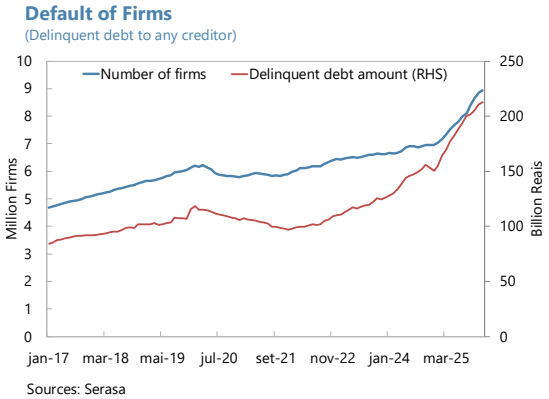
Growth has moderated with tight monetary policy, a scaling back of fiscal support, and heightened global uncertainty.



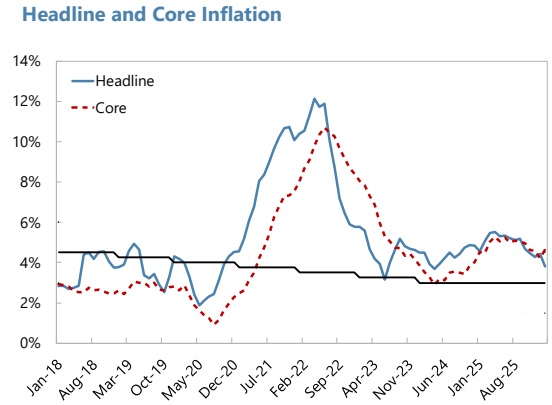
… after a new tightening cycle in response to a flare-up in inflation 2024.



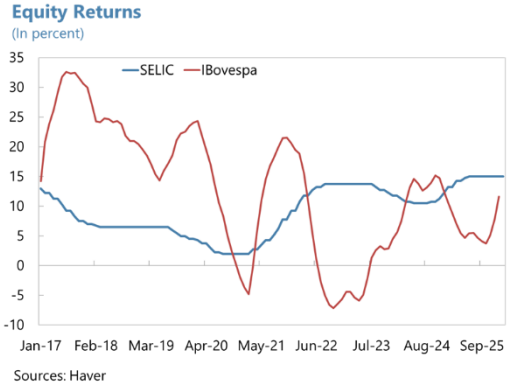
Default frequency of firms has gone up significantly in 2025 amid high interest rates.



Inflation began to decline in 2025…



… issued by large companies in utilities and services sectors.



Judicial reorganizations have gone up, with number of companies involved surpassing that in 2015-16 crisis.

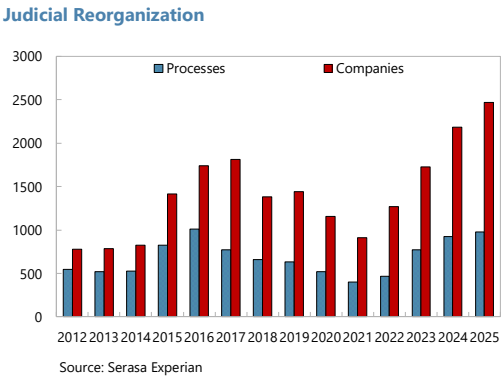
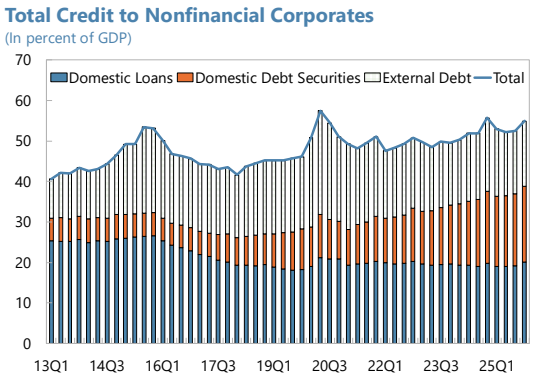
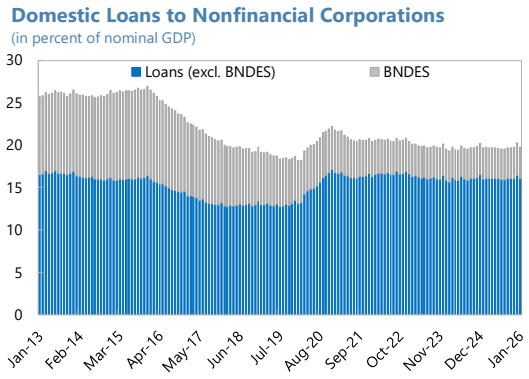


Figure 4. Brazil: Nonfinancial Corporates Financing

Domestic loans to firms declined after the BNDES reform while bond financing has increased significantly.

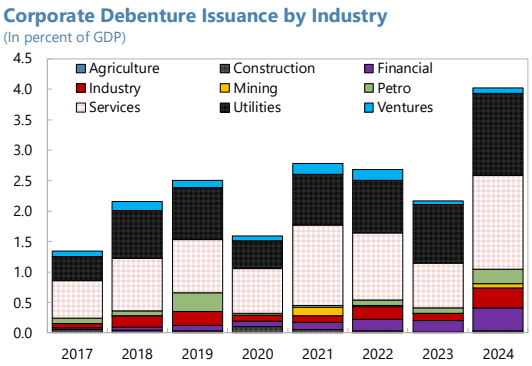


Loans from domestic financial institutions declined mainly due to lower financing from BNDES post-reform.



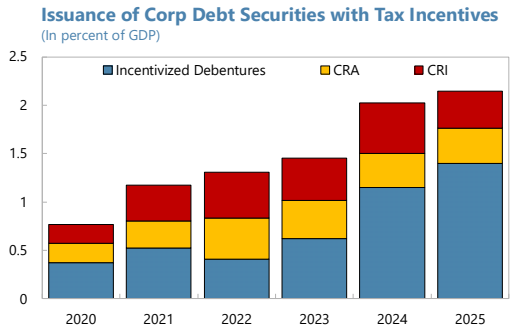
Sources: Haver

Large firms, esp. those in utilities and services, increased bond issuance…



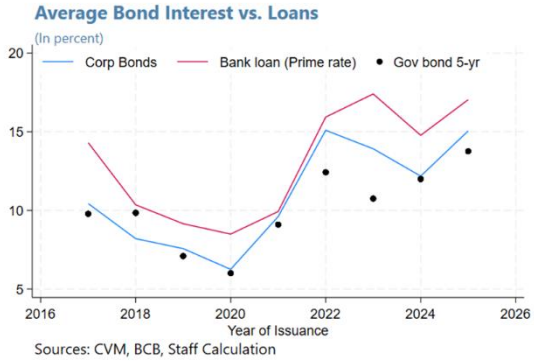
Sources: CVM and Staff Calculation

…with growing issuance of debt securities with tax incentives.



Note: CRA and CRI are fixed-income asset-backed securites used to fund agriculture and real estate sectors, respectively.

Debt securities with tax incentives are attractive to investors…



Sources: CVM, BCB, Staff Calculation

…leading to narrower spread compared to government bonds.

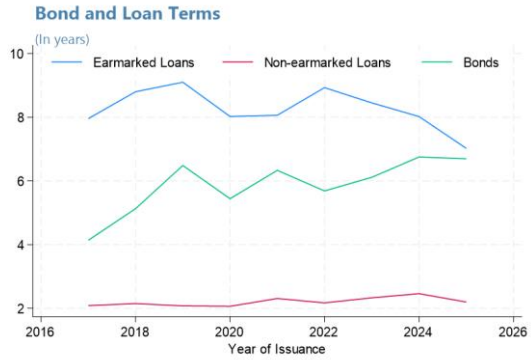
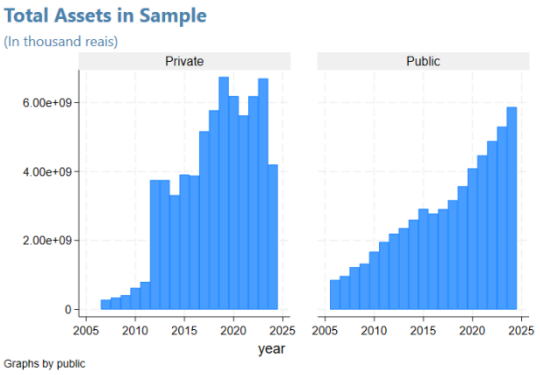
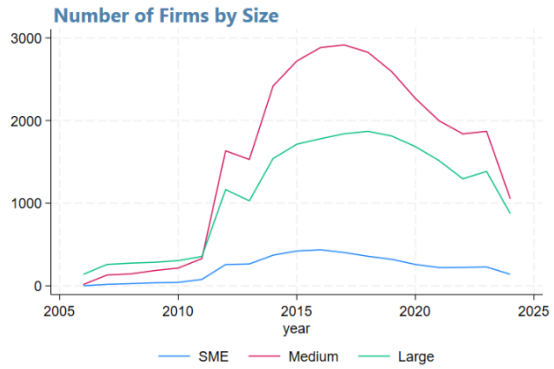


Figure 5. Brazil: Capital IQ Sample

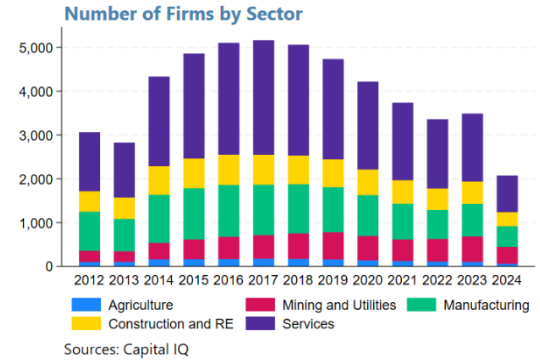
Sample coverage was good after 2012



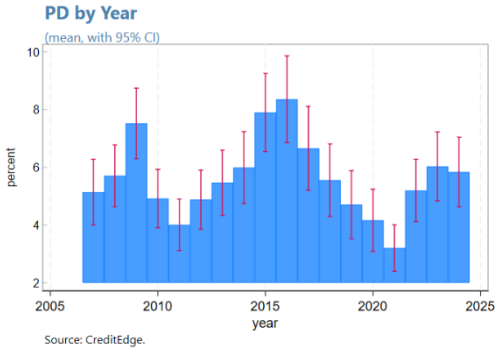
About half of the firms are medium-sized and the coverage of SMEs is low.



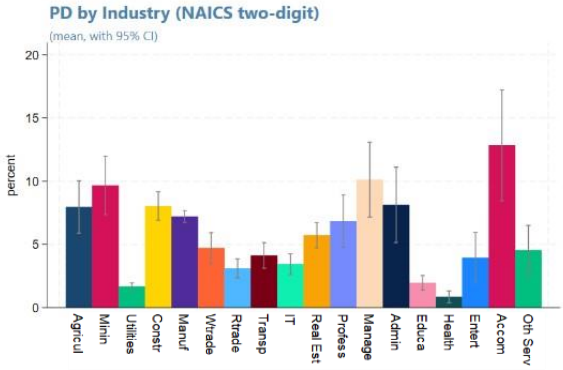
The sample covers services, manufacturing, and all major industries.



PD in the CreditEdge sample fluctuates with business cycles.



PD also varies across industries, with highest risk in accommodation and food services.



LGD is between 40-60 percent, with similar estimates between Capital IQ and BCB.

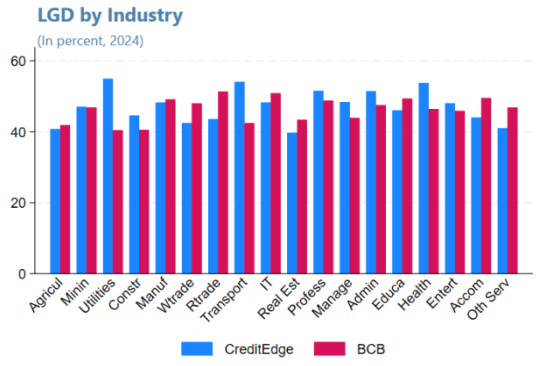
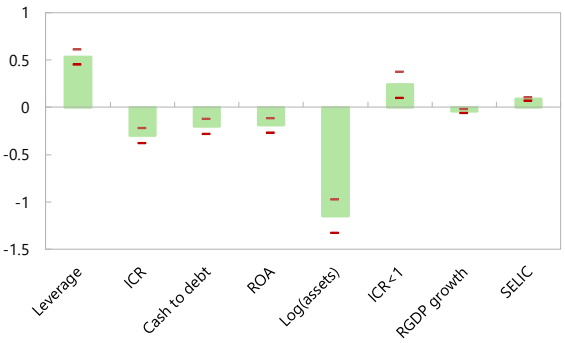


Figure 6. Brazil: PD and LGD Models

All coefficients are significantly different from zero and take the expected signs.

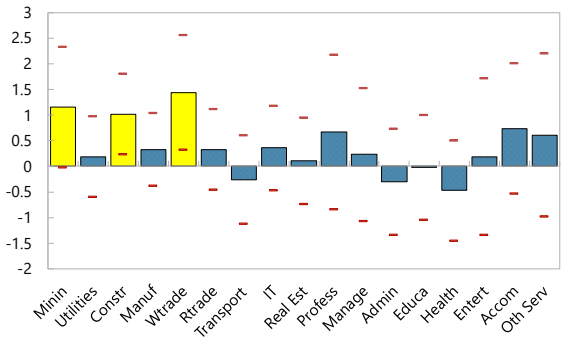
PD Model Coefficients



Note: the first five variables are normalized.

Companies in mining, construction, and wholesale trade have significantly higher PD than those in agriculture

Industry fixed effects

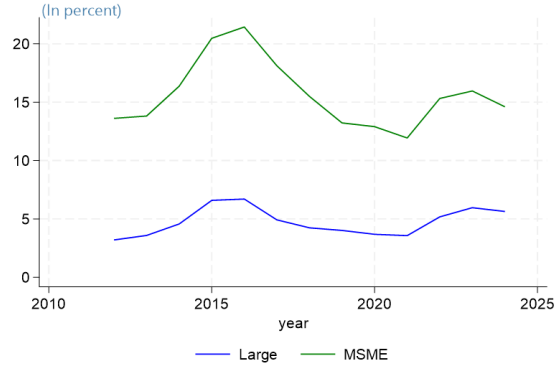


Note: Agriculture is the reference industry.

Imputed PD among MSMEs is much higher than large firms, with levels …

PD in Capital IQ Sample

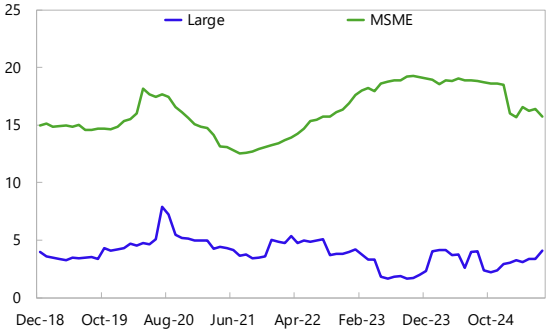
(In percent)



… with levels and trends similar to PD calculated by the BCB using loan level data.

PD in BCB Data

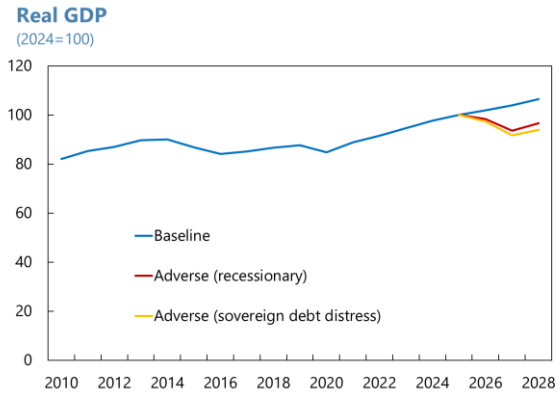
(In percent)



Sources: Financial Stability Report (BCB)

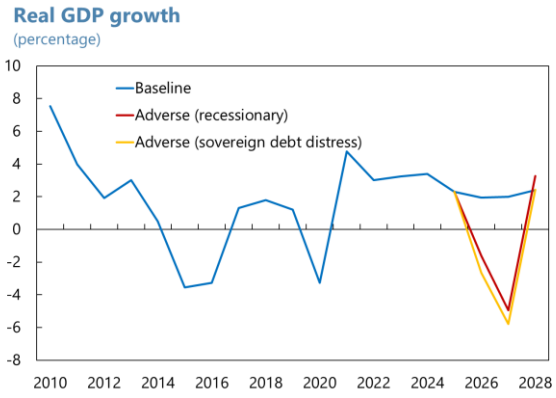
Figure 7. Brazil: Macro-Financial Scenarios

Sovereign debt distress scenario adds a fiscal confidence crisis on the recessionary scenario, with sharper GDP drop



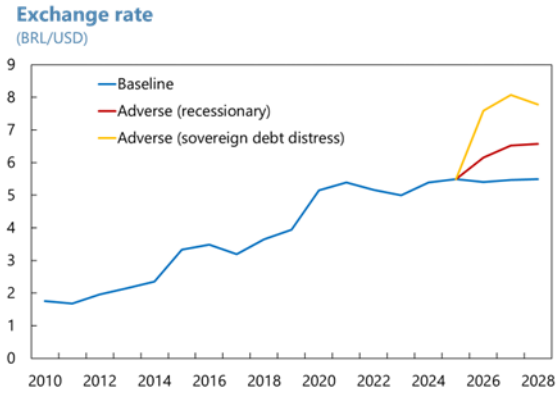
Source: Haver and IMF staff calculations.

Real GDP contracts more in sovereign distress scenario.



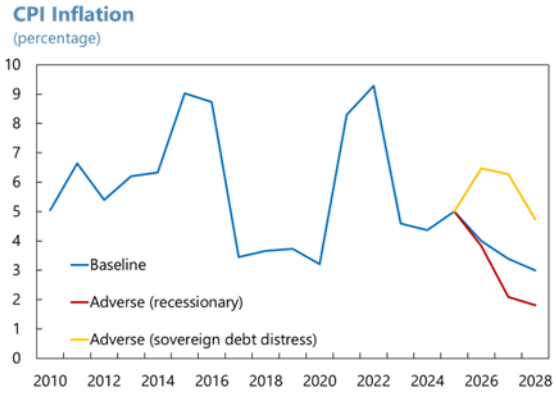
Source: Haver and IMF staff calculations.

The confidence crisis leads to sharper FX depreciation…



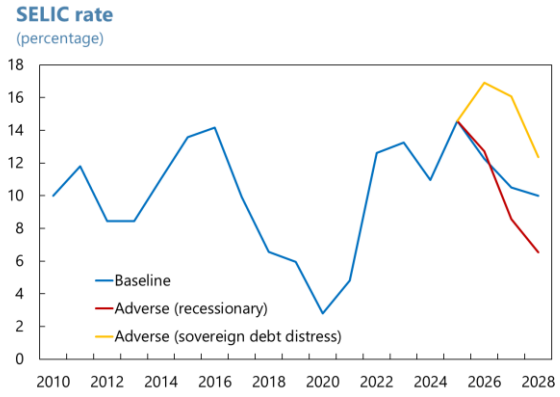
Source: Haver and IMF staff calculations.

… contributing to higher inflation despite economic contraction



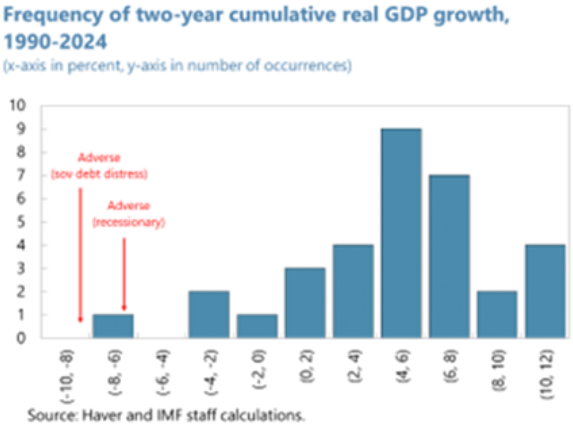
Source: Haver and IMF staff calculations.

The central bank cuts rates under recessionary scenario but hikes rates to combat inflation in the confidence crisis.



Source: Haver and IMF staff calculations.

The scenario is severe with real GDP contraction 3.1 standard deviation below average since 1990



Source: Haver and IMF staff calculations.

Figure 8. Brazil: NFC Stress Testing Results
(By Firm Size)

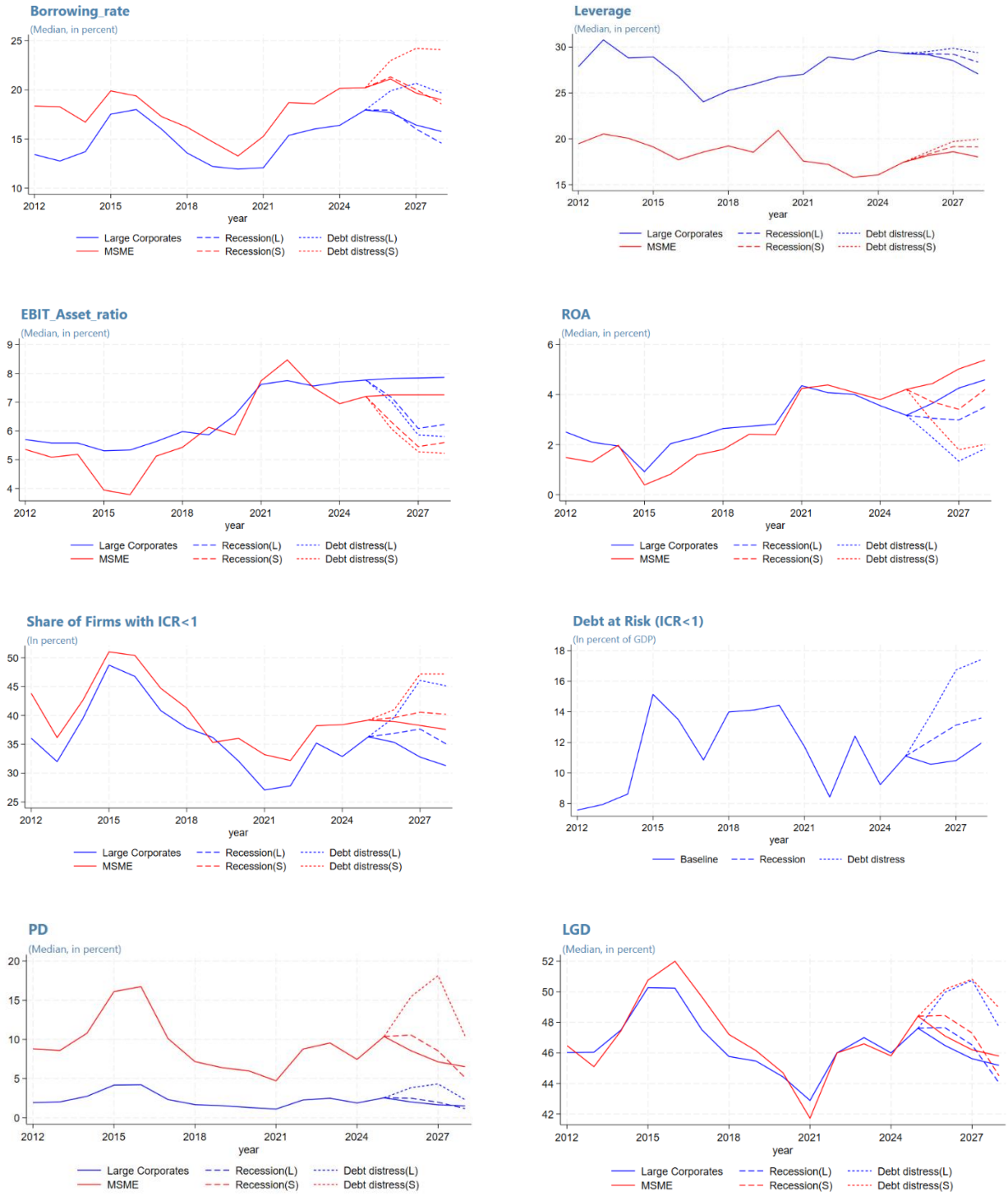


Figure 9. Brazil: NFC Stress Testing Results
(By Economic Sectors)

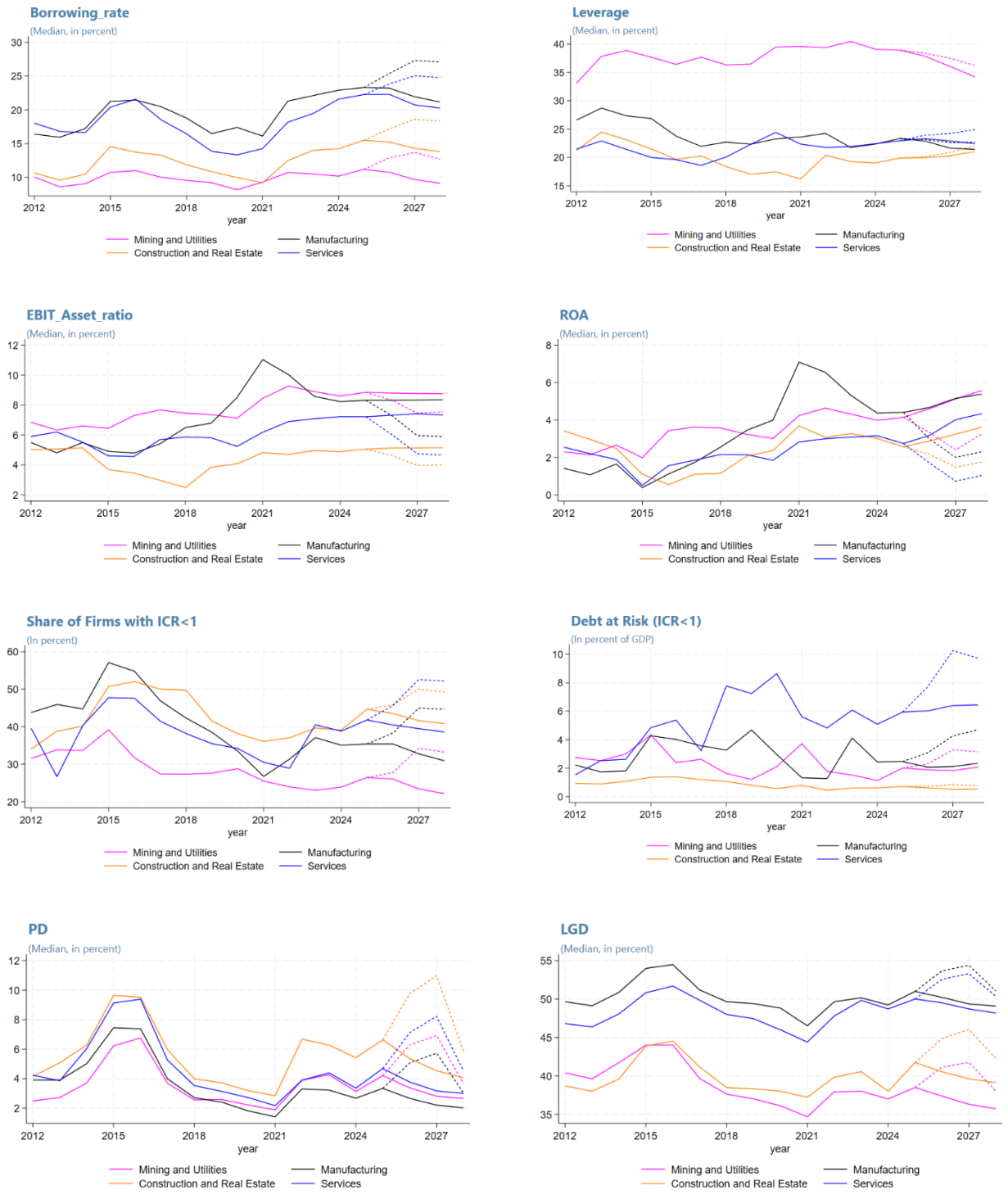
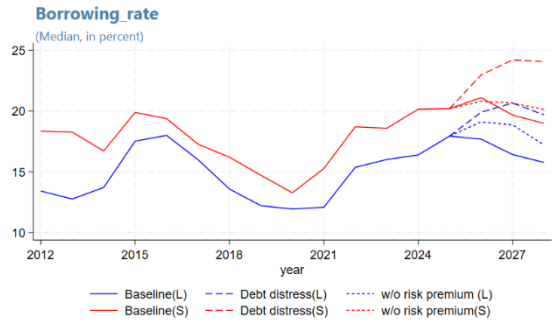
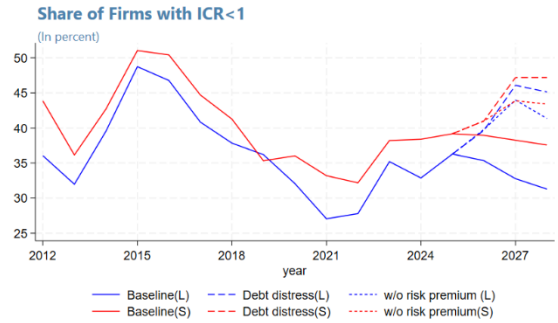


Figure 10. Brazil: The Role of Risk Premia

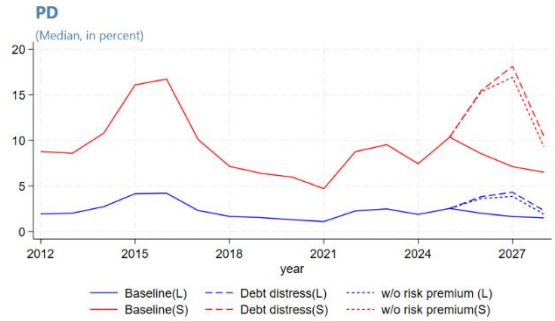
Risk premia contribute to over half of the rise in interest rates under the debt distress scenario, with a greater increase observed for MSMEs



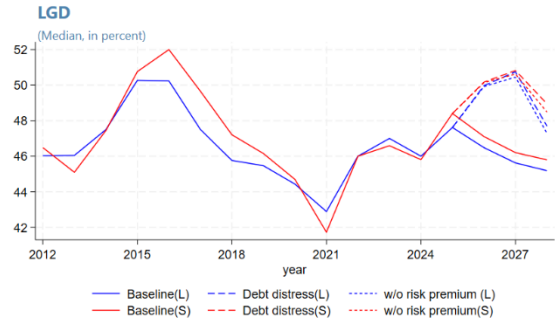
Ignoring risk premium would underestimate the number of firms with ICR<1 under the adverse scenario.



Adding the risk premium also increases firms' PD, but the impact is modest since PD depends on various factors.



The same is true for LGD.



Annex I. Data Collection and Processing

Firm-level balance sheet and cash flow data are downloaded from S&P Capital IQ pro. Firm-level default probability (PD) and loss given default (LGD) are obtained from CreditEdge of Moody's analytics. This annex describes the process of downloading and cleaning the data for analysis.

Downloading Data from Capital IQ Pro

Capital IQ Pro is a new and improved version of the Capital IQ database from Standard & Poor's (S&P). Compared to the previous version, it covers a broader sample of private companies with more timely updates on balance sheets and cash flow information. In the case of Brazil, companies covered by Capital IQ pro accounts for 75% of total corporate debt and more than one-third of the value added in the whole economy.

We use the "Screener" feature to extract balance sheet information of companies in Brazil. Capital IQ contains millions of companies around the world, and so the user needs to narrow down the sample to build a dataset that is not too big to be downloaded in excel. We use the following criteria to build our dataset: 1) geographical area is set to Brazil; 2) key accounting variables is not "NA" (e.g., total revenues, total assets, interest expenses, cost of goods sold, and EBIT). After applying these screening criteria, the sample size is around 10,000 firms. Choose "Display columns" to select variables.

Next we select the variables and years for the sample. Capital IQ contains hundreds of variables from multiple sources. Given the size limit imposed by excel when downloading the data, it is important to be parsimonious in choosing the variables and years. For our purpose, we include company identifiers, industry classification, parent company information, and key accounting variables from S&P Capital IQ Fundamentals covering 2006-2024. The full list of variables is detailed in Table A1.[12]

Table A1. List of Variables

Company information	Accounting variables
Company name	Total revenue
ID (capital IQ specific)	Total assets
ISIN (International Securities Identification Number)	Total debt
NAICS (North American Industry Classification System)	Total equity
Capital IQ Industry Classification	Cash and equivalents
NACE (European classification of business activities)	Cash and short-term investment
Status	Interest Expenses
Type (public or private)	Interest income
Parent company name	Cost of Goods Sold (COGS)

[12] If the full dataset is too big to be downloaded into one excel file, consider splitting the sample by variables, e.g. eight variables per dataset, and use the company ID to merge the dataset when processing the data. New variables can be easily added.

Parent company ID	EBIT
Parent exchange ticker	EBT
Publicly listed indicator	EBITDA
	Current Taxes
	Net income
	Capital Expenditure
	Net Property, Plant, and Equipment

PDs and LGDs from CreditEdge

We obtain PDs and LGDs of public companies from Moody's KMV CreditEdge. The PD is referred to as the Expected Default Frequency (EDF) in CreditEdge, over a one-year horizon. LGD is proxied by Sector LGD in CreditEdge imputed from spreads. Unlike the EDF which is firm-specific, the sector LGD is a generic measure for a thirteen sector three region two seniority matrix. The measure is updated daily. The LGD data can be complemented by LGD estimates for each industry provided by the central bank in the context of stress testing.

Data Cleaning

The first step is to merge the PD and LGD with the Capital IQ sample. This can be done by matching the Issuer ISIN of the publicly listed firms.
Next, we address potential duplication when both subsidiaries and parent companies are in the sample. Using information on parent company's ID for each subsidiary, we drop the subsidiary when its parent company is also in the sample.

We exclude firms in the financial sector from the corporate stress testing analysis. This reflects the fundamental difference between financial firms and nonfinancial corporates in terms of business models and risk metrics. The solvency and liquidity of financial firms, including banks and NBFIs, are usually analyzed in detail in the IMF's FSAP.

We eliminate problematic observations and outliers following standard procedures. For variables that should take non-negative values, such as total revenues, total assets, cash and equivalents, and cost of goods sold, we drop observations with negative values. For firms with no interest expenses, we assign a small number so that the interest coverage ratio would not be missing for the estimation of the PD equation. For the accounting ratios included in the PD equation, we trimmed the top and bottom five percent of the values to reduce the influence of outliers on the estimation.



PUBLICATIONS

Corporate Sector Stress Testing with Endogenous Balance Sheet Dynamics
Working Paper No. WP/2026/209